Import Data

In [1]:
```
import pandas as pd
import seaborn as sns
import matplotlib.pyplot as plt
%matplotlib inline
import numpy as np
```

In [2]:
```
df = pd.read_csv(r"D:\Downloads\Elevvo_Projects\Titanic-Dataset.csv")
```

In [3]:
```
df
```

Out[3]:

	PassengerId	Survived	Pclass	Name	Sex	Age	SibSp	Parch	Ticket	F
0	1	0	3	Braund, Mr. Owen Harris	male	22.0	1	0	A/5 21171	7.2
1	2	1	1	Cumings, Mrs. John Bradley (Florence Briggs Th...	female	38.0	1	0	PC 17599	71.2
2	3	1	3	Heikkinen, Miss. Laina	female	26.0	0	0	STON/O2. 3101282	7.9
3	4	1	1	Futrelle, Mrs. Jacques Heath (Lily May Peel)	female	35.0	1	0	113803	53.1
4	5	0	3	Allen, Mr. William Henry	male	35.0	0	0	373450	8.0
...	...	...	...	...	...	...	...	...	...	
886	887	0	2	Montvila, Rev. Juozas	male	27.0	0	0	211536	13.0
887	888	1	1	Graham, Miss. Margaret Edith	female	19.0	0	0	112053	30.0
888	889	0	3	Johnston, Miss. Catherine Helen "Carrie"	female	NaN	1	2	W./C. 6607	23.4
889	890	1	1	Behr, Mr. Karl Howell	male	26.0	0	0	111369	30.0
890	891	0	3	Dooley, Mr. Patrick	male	32.0	0	0	370376	7.7

891 rows × 12 columns

In [4]:
```
df.info()
```

```
<class 'pandas.core.frame.DataFrame'>
RangeIndex: 891 entries, 0 to 890
Data columns (total 12 columns):
 #   Column       Non-Null Count  Dtype
---  ------       --------------  -----
 0   PassengerId  891 non-null    int64
 1   Survived     891 non-null    int64
 2   Pclass       891 non-null    int64
 3   Name         891 non-null    object
 4   Sex          891 non-null    object
 5   Age          714 non-null    float64
 6   SibSp        891 non-null    int64
 7   Parch        891 non-null    int64
 8   Ticket       891 non-null    object
 9   Fare         891 non-null    float64
 10  Cabin        204 non-null    object
 11  Embarked     889 non-null    object
dtypes: float64(2), int64(5), object(5)
memory usage: 83.7+ KB
```

In [5]:
```
df.describe()
```

Out[5]:

	PassengerId	Survived	Pclass	Age	SibSp	Parch	Fare
count	891.000000	891.000000	891.000000	714.000000	891.000000	891.000000	891.000000
mean	446.000000	0.383838	2.308642	29.699118	0.523008	0.381594	32.204208
std	257.353842	0.486592	0.836071	14.526497	1.102743	0.806057	49.693429
min	1.000000	0.000000	1.000000	0.420000	0.000000	0.000000	0.000000
25%	223.500000	0.000000	2.000000	20.125000	0.000000	0.000000	7.910400
50%	446.000000	0.000000	3.000000	28.000000	0.000000	0.000000	14.454200
75%	668.500000	1.000000	3.000000	38.000000	1.000000	0.000000	31.000000
max	891.000000	1.000000	3.000000	80.000000	8.000000	6.000000	512.329200

Clean Data

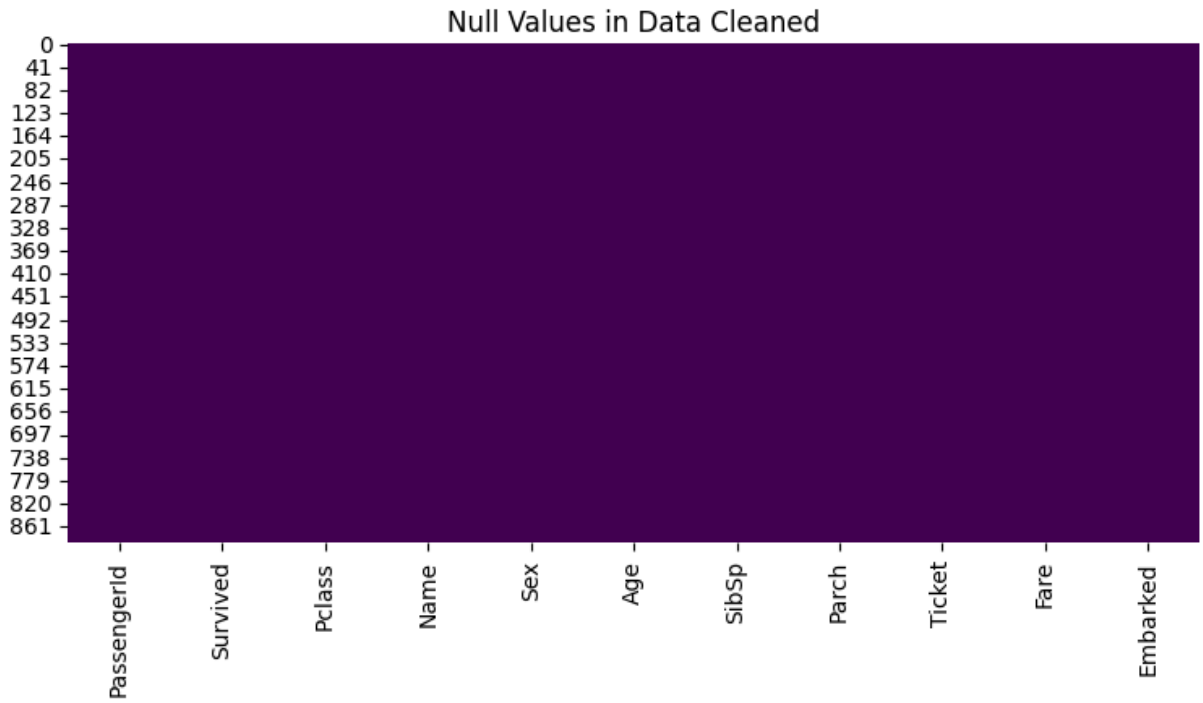
In [6]:
```
df['Age'] = df['Age'].fillna(df['Age'].median)              # Filling the blank ro
```

In [7]:
```
df['Embarked'] = df['Embarked'].fillna(df['Embarked'].mode()[0])             # Fil
```

In [8]:
```
df.drop(columns= ['Cabin'],inplace=True)
```

Checking Null after cleaning

In [9]:
```
plt.figure(figsize=(9,4))
sns.heatmap(df.isnull(),cbar= False, cmap = 'viridis')
plt.title('Null Values in Data Cleaned')
plt.show();
```



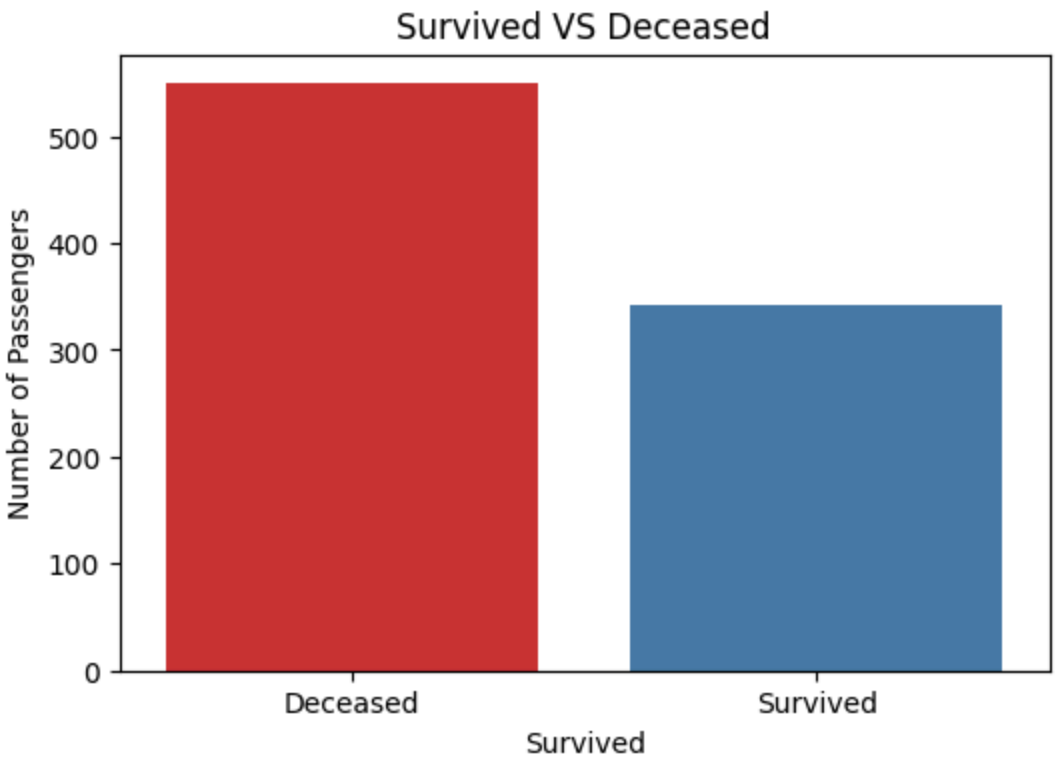
How much people survived?

In [10]:
```
df['Survived'].value_counts()
```

Out[10]:
```
Survived
0    549
1    342
Name: count, dtype: int64
```

In [11]:
```
plt.figure(figsize=(6,4))
sns.countplot(x="Survived", data = df, palette = 'Set1')
plt.xticks([0,1],['Deceased','Survived'])
plt.title('Survived VS Deceased')
plt.ylabel('Number of Passengers')
plt.show();
```

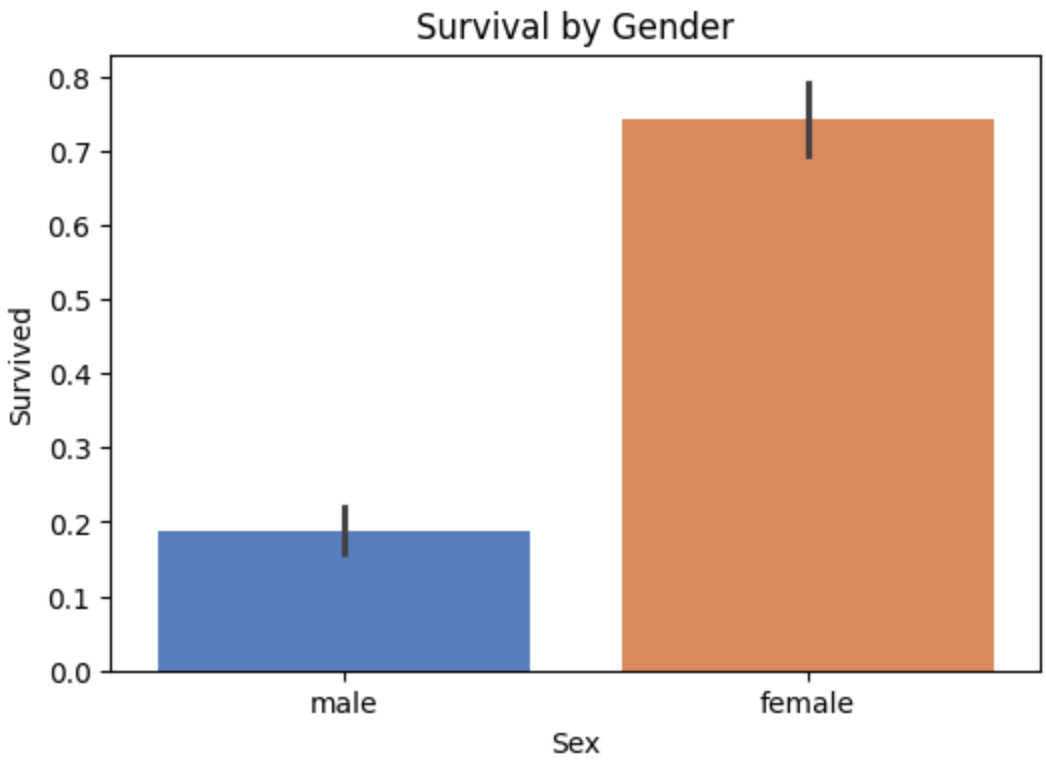
```
C:\Users\hmmuz\AppData\Local\Temp\ipykernel_15612\3245550497.py:2: FutureWarning:

Passing `palette` without assigning `hue` is deprecated and will be removed in v0.1
4.0. Assign the `x` variable to `hue` and set `legend=False` for the same effect.

  sns.countplot(x="Survived", data = df, palette = 'Set1')
```



Survival by Gender

In [20]:
```
g_survival = df.groupby('Sex')['Survived'].mean()
plt.figure(figsize=(6,4))
sns.barplot(x = 'Sex', y = 'Survived', data = df, palette = 'muted')
plt.title('Survival by Gender')
plt.show();
```

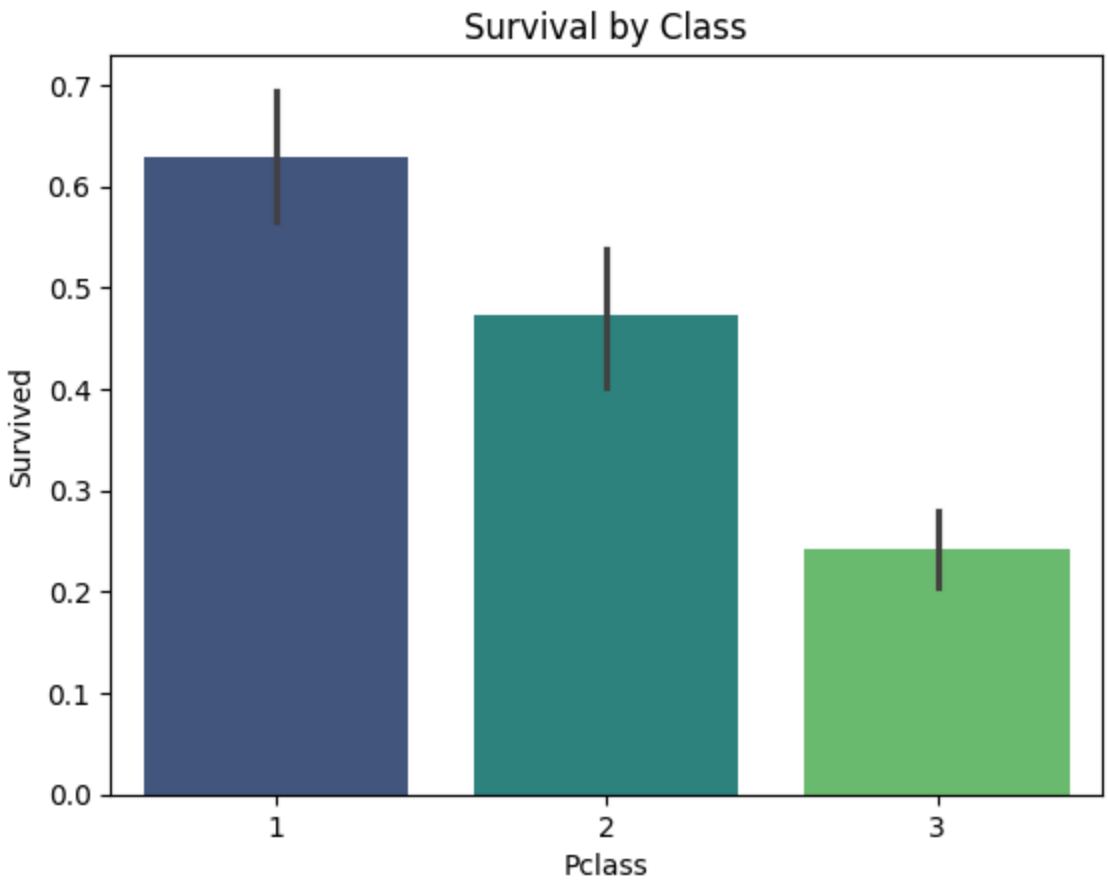
```
C:\Users\hmmuz\AppData\Local\Temp\ipykernel_15612\3279120021.py:3: FutureWarning:

Passing `palette` without assigning `hue` is deprecated and will be removed in v0.1
4.0. Assign the `x` variable to `hue` and set `legend=False` for the same effect.

  sns.barplot(x = 'Sex', y = 'Survived', data = df, palette = 'muted')
```



Survival by Class

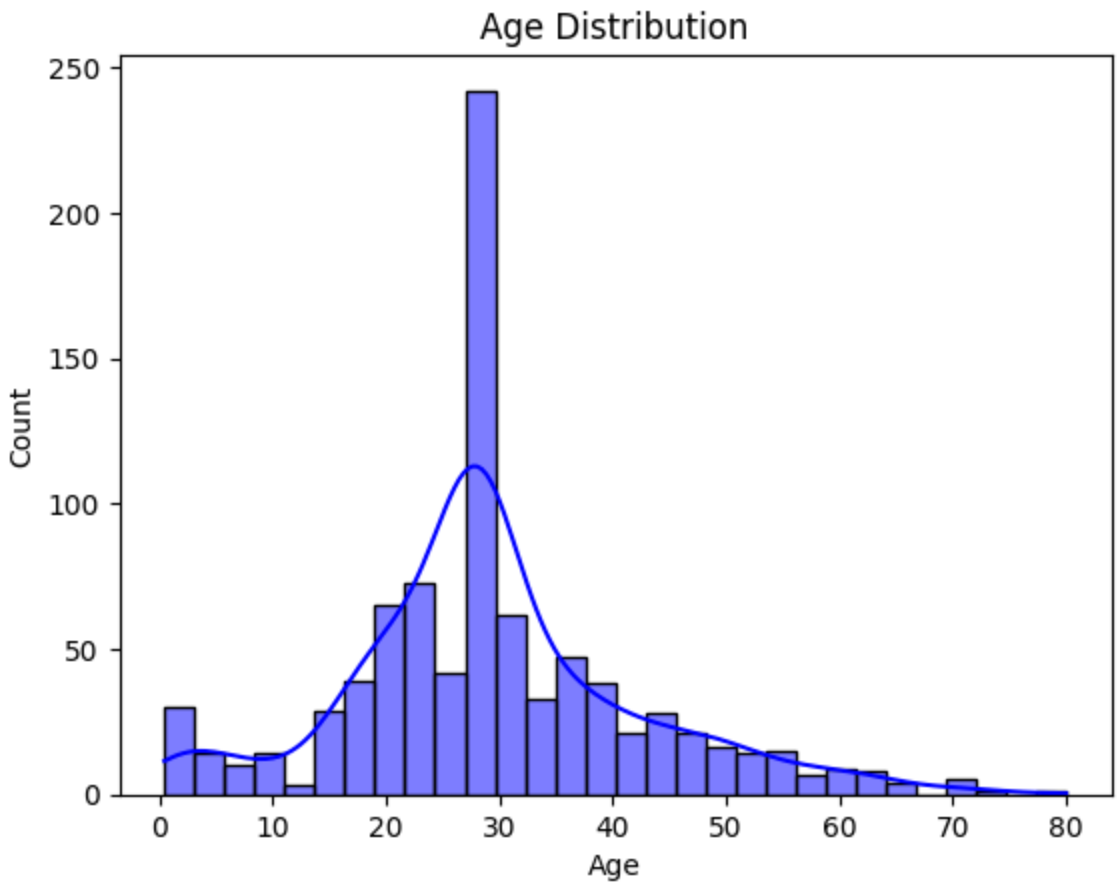
In [22]:
```
df.groupby('Pclass')['Survived'].mean()
sns.barplot(x ='Pclass',y ='Survived', data = df, palette = 'viridis')
plt.title('Survival by Class')
plt.show();
```

```
C:\Users\hmmuz\AppData\Local\Temp\ipykernel_15612\4091698362.py:2: FutureWarning:

Passing `palette` without assigning `hue` is deprecated and will be removed in v0.1
4.0. Assign the `x` variable to `hue` and set `legend=False` for the same effect.

  sns.barplot(x ='Pclass',y ='Survived', data = df, palette = 'viridis')
```



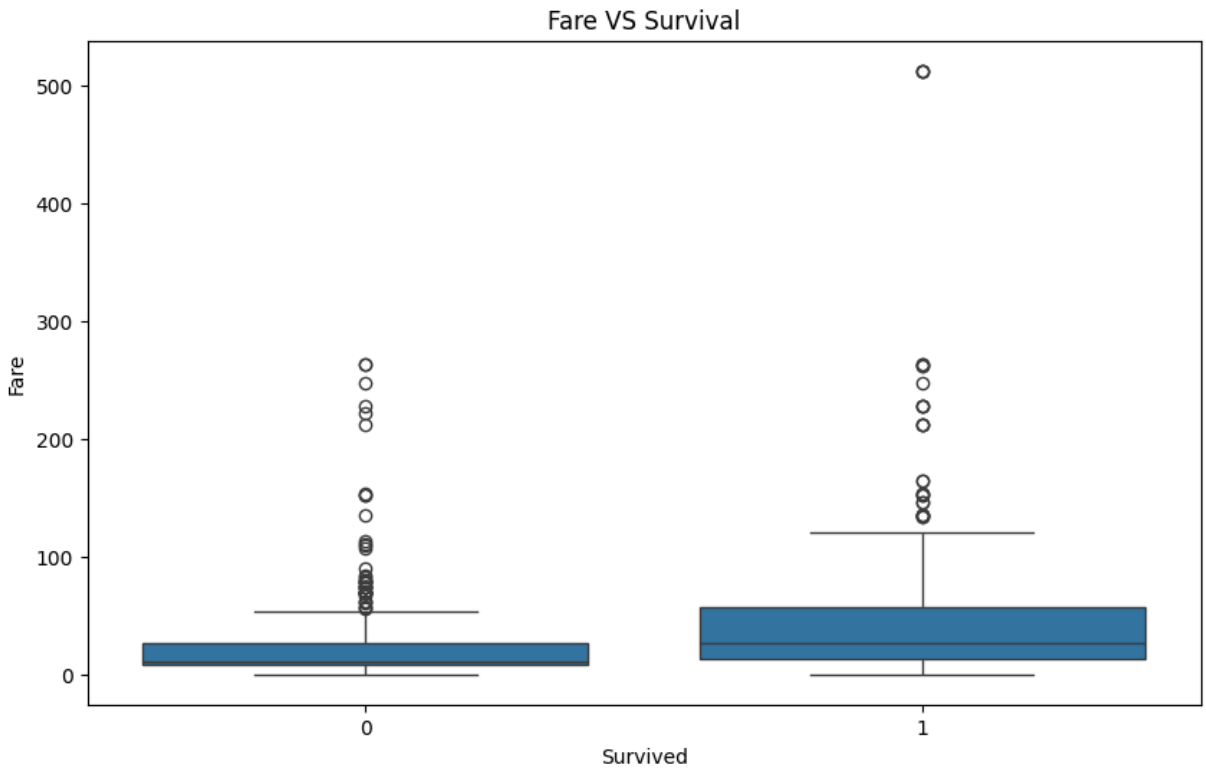
Age distribution

In [28]:
```
df['Age'] = pd.to_numeric(df['Age'], errors='coerce')
df['Age'] = df['Age'].fillna(df['Age'].median())
sns.histplot(data=df, x='Age', bins= 30,kde=True, color = 'blue')
plt.title('Age Distribution')
plt.show();
```



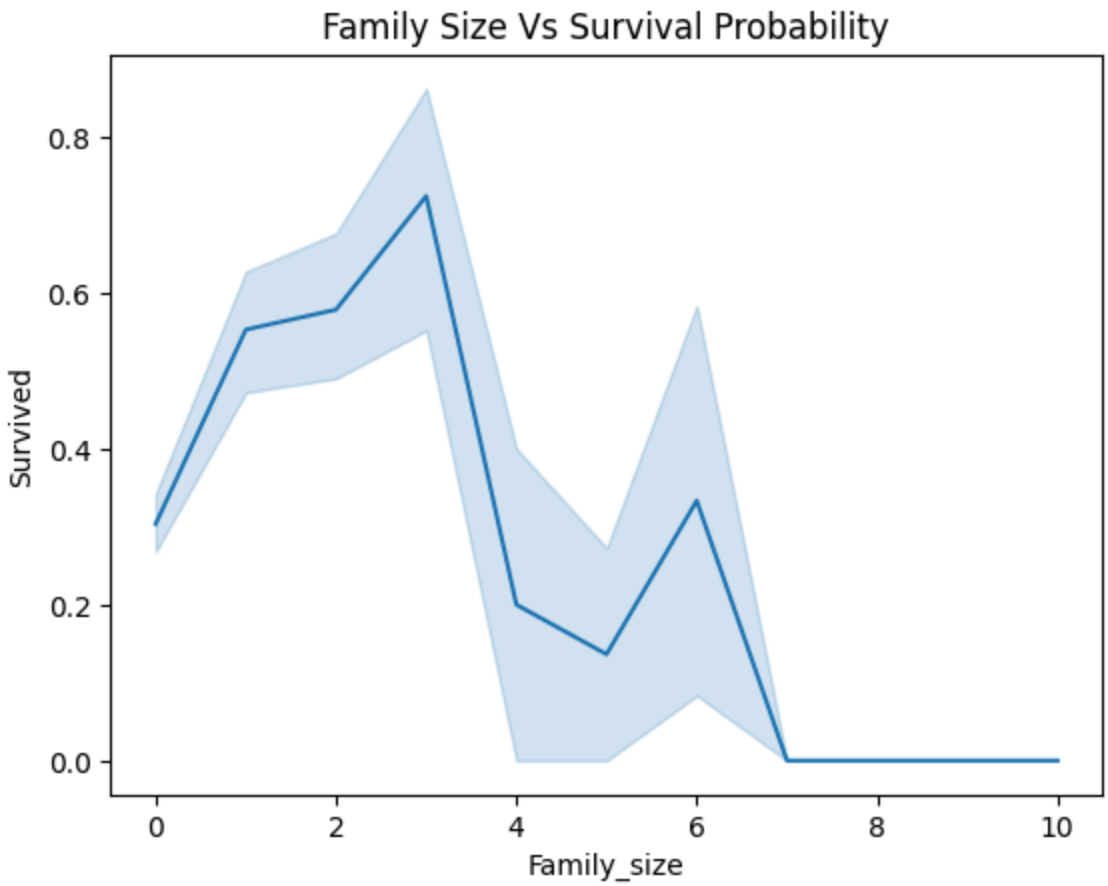
Fare Vs Survival

In [29]:
```
plt.figure(figsize=(10,6))
sns.boxplot(x='Survived', y = 'Fare', data = df)
plt.title('Fare VS Survival')
plt.show();
```



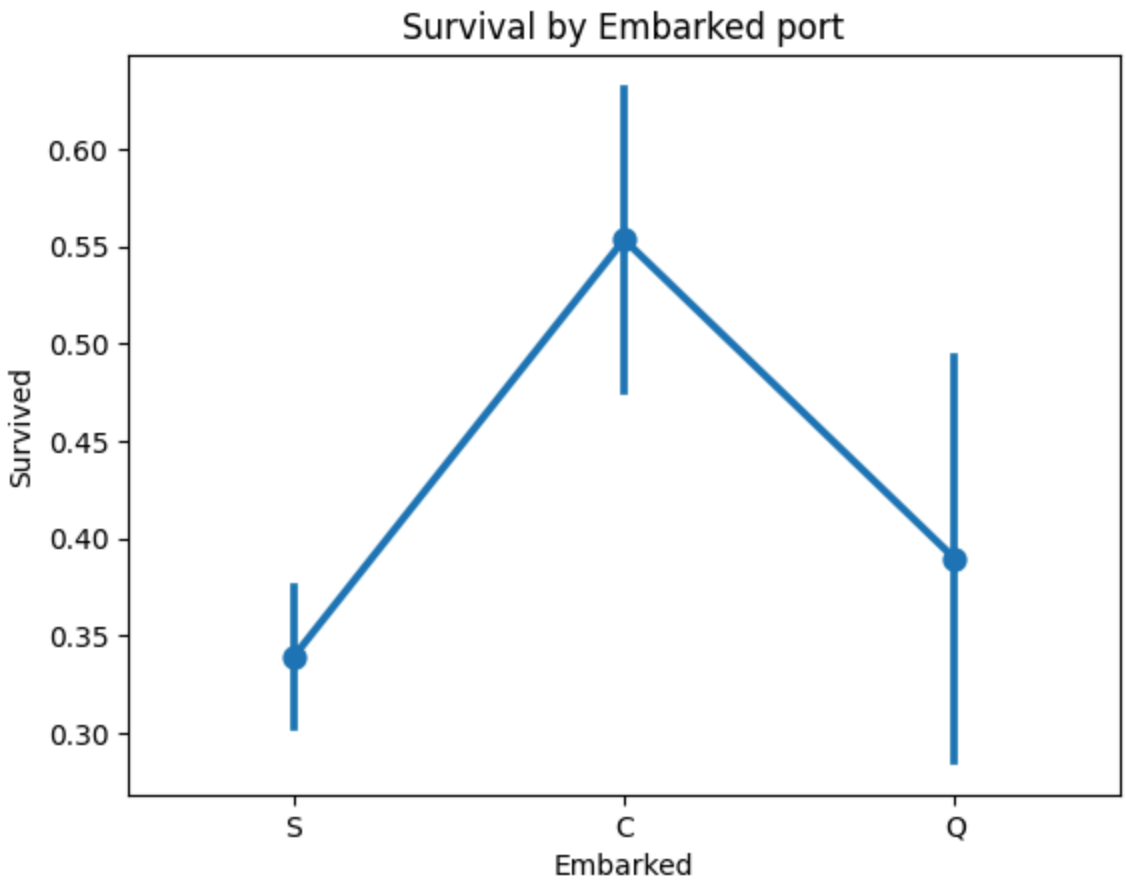
Family size VS Survival

In [31]:
```
df['Family_size'] = df['SibSp']+df['Parch']
sns.lineplot(x='Family_size',y='Survived',data = df)
plt.title('Family Size Vs Survival Probability')
plt.show();
```



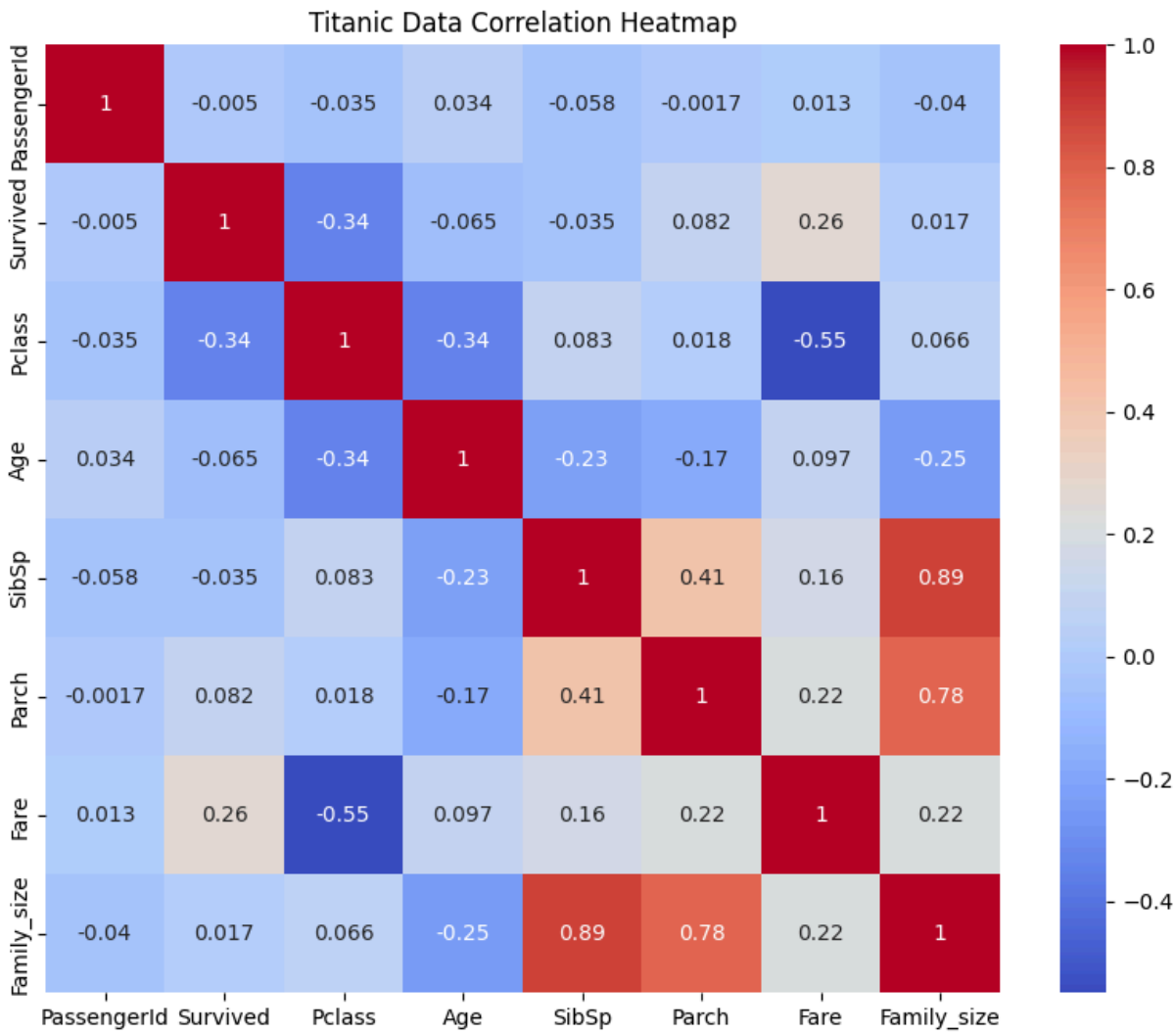
Embarked impact on Survival

In [34]:
```
sns.pointplot(x='Embarked',y='Survived',data=df)
plt.title('Survival by Embarked port')
plt.show();
```



Find Numeric Correlations

In [37]:
```
numeric_data = df.select_dtypes(include=['float64', 'int64'])
plt.figure(figsize=(10, 8))
sns.heatmap(numeric_data.corr(), annot=True, cmap='coolwarm')
plt.title('Titanic Data Correlation Heatmap')
plt.show()
```



Prepare ML model

In [38]:
```
from sklearn.model_selection import train_test_split
from sklearn.linear_model import LogisticRegression
from sklearn.metrics import accuracy_score, confusion_matrix
```

Feature Selection & Encoding

In [40]:
```
df_ml = df[['Survived', 'Pclass', 'Sex', 'Age', 'SibSp', 'Parch', 'Fare', 'Embarked
df_ml = pd.get_dummies(df_ml, columns=['Sex', 'Embarked'], drop_first=True)
```

In [41]:
```
X = df_ml.drop('Survived', axis=1) # Input data
y = df_ml['Survived']              # Target (0 or 1)
```

Train-Test Split

In [42]:
```
X_train, X_test, y_train, y_test = train_test_split(X, y, test_size=0.2, random_sta
```

Model Training

In [43]:
```
model = LogisticRegression(max_iter=1000)
model.fit(X_train, y_train)
```

Out[43]:
```
▾ LogisticRegression
▶ Parameters
```

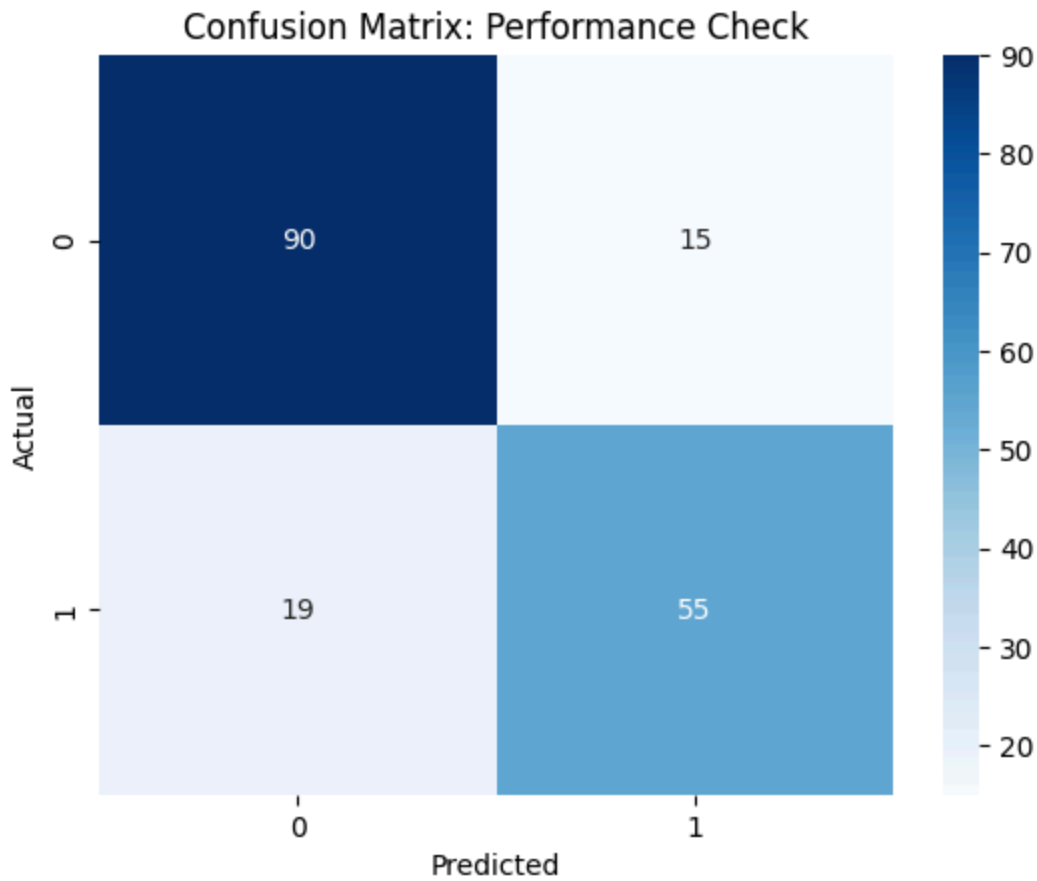
Predictions & Accuracy

In [44]:
```
predictions = model.predict(X_test)
score = accuracy_score(y_test, predictions)

print(f"Model Accuracy: {score * 100:.2f}%")
```

```
Model Accuracy: 81.01%
```

Confusion Matrix

In [45]:
```
sns.heatmap(confusion_matrix(y_test, predictions), annot=True, fmt='d', cmap='Blues
plt.xlabel('Predicted')
plt.ylabel('Actual')
plt.title('Confusion Matrix: Performance Check')
plt.show()
```



The confusion matrix reveals that the Logistic Regression model is particularly strong at identifying deceased passengers, though it occasionally misclassifies survivors due to the high variance in the 'Age' and 'Fare' features.

In [ ]: